

Experiment no.

Aim: Determination of magnetizing characteristics of a separately excited DC generator at rated speed.

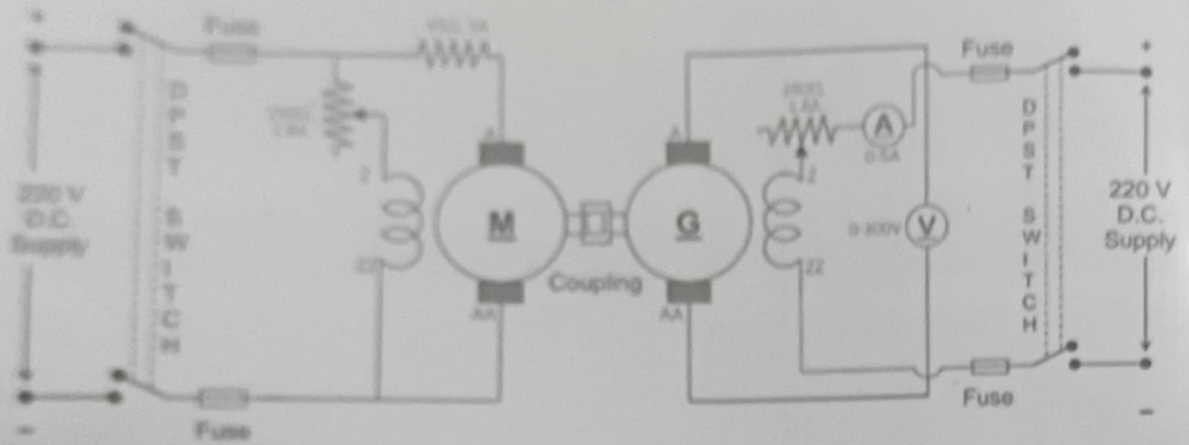
Instruments Required:

Sl.no.	Name	Type	Range	Quantity
1	Voltmeter	MC	0-300 V	1 no.
2	Ammeter	MC	0-5 A	1 no.
3	Rheostat	Single Tube	45 ohm, 5 A	1 no.
4	Rheostat	Single Tube	290 ohm, 1.8 A	1 no.
5	Tachometer	Digital	0-2000 RPM	1 no.

Specification of machine

	DC shunt motor	DC shunt generator
Capacity	5 HP	3 KW
Rated voltage	220V	220V
Rated current	17S	20.5A
Rated speed	1500 RPM	1500 RPM

Theory: The magnetization characteristics or the open circuit characteristics (OCC) of a generator shows the relation between the induced EMF at no-load and the field current.



Circuit Diagram

Observation Table:

S.no	Field Current (A)	Induced EMF (V)	Speed (RPM)

The EMF induced in the armature of a DC generator is given by

$$E_g = \frac{P\phi ZN}{60A}$$

Where

ϕ is the useful flux per pole in Wb

Z is the total number of armature conductor

P is the number of poles in a generator

A is the number of parallel lanes within the armature

N is the rotation of armature in RPM

When there is no current is flowing in the field winding, the EMF induced due to the residual flux present in the machine. If the field current is slowly increased the flux first increases linearly resulting in the linear increase in the EMF but after a certain value of the field current, the saturation of magnetism causes a slower increase in the field current resulting in the bending of magnetization curve.

Procedure:

1. Connections are made as per the circuit diagram.
2. With maximum resistance in the armature circuit and minimum resistance in the field circuit of the motor, supply is switched on.
3. The resistance in the armature circuit is slowly cut out.
4. As motor picks up the speed, the rated speed of the motor is obtained constant throughout experiment.
5. The voltage reading is recorded corresponding to the zero field current in the DC generator.
6. DC supply is given to the potential divider circuit arrangement with minimum current flowing the generator field winding. This current will slowly increase with moving of jockey potential divider in one direction.
7. Only the reading of voltmeter and ammeter are recorded till all voltmeter reads 125% of rated voltage of generator.

Graph: Plot graph between field current and induced emf.

Result:

Precautions:

1. Do not touch any terminal or switch without ensuring that it is dead.
2. Never work alone in the laboratory.
3. Make sure that all the connections are right, before switching on any circuit, faulty connections may cause drawing large power resulting in damage of parts of instruments.
4. The circuit should be de-energized before changing any connections.